CAEA: Supplementary Proofs and Reproduction Details

A Proofs

A.1 Proof of Proposition 1

For any feasible π, its class marginal is b. The KL chain rule gives

$$D_{\mathrm{KL}}(\pi\|\pi_P) = D_{\mathrm{KL}}(b\|m_P) + \sum_c b(c) D_{\mathrm{KL}}\big(\pi(\cdot \mid c)\|\pi_P(\cdot \mid c)\big). \tag{1}$$

Every conditional term is nonnegative and is zero exactly when the conditional distributions agree on positive-mass classes. This yields the unique full policy π_b. Classes of zero target mass carry no response probability, so their arbitrary conditional representation does not affect uniqueness of the policy. Taking logs in Equation (1) gives $\log \pi_P = \log \pi_0 + r/\beta - \log Z$, hence $J(\pi) = \beta \log Z - \beta D_{\mathrm{KL}}(\pi\|\pi_P)$. Subtracting its value at π_P proves Equation (6). The cost is zero exactly when $b = m_P$; for full repair $b = \tau_x$. All quantities are finite by the finite response space, bounded reward, positive reference, and feasible target support. The same assumptions ensure existence of the stated optimum. □

A.2 Proof of Theorem 2

The reward difference is $r(1, s) - r(0, s) = k(2s - 1)$. A uniform reference therefore gives Equation (7). Since $\mathrm{sigmoid}(-v) = 1 - \mathrm{sigmoid}(v)$, $m_a(1 \mid 1 - s) = 1 - m_a(1 \mid s)$. A symmetric distribution of s thus has pooled report probability $1/2$. Mixture with the uniform target preserves that symmetry. Because the outcome is independent of the stance and response-sampling randomness conditional on the fixed evidence, the joint pooled law of (Z, q) is the product of uniform Bernoulli laws. Every statistic determined by that law has the same distribution, including expected Brier loss $\Pr(q \neq Z) = 1/2$ and regret $\mathbb{E}(q - 1/2)^2 = 1/4$.

Differentiating the mean-response update yields

$$F'_\lambda(s) = 1 - \eta + 2\eta(1 - \lambda)a\,\mathrm{sigmoid}(a(2s - 1))[1 - \mathrm{sigmoid}(a(2s - 1))]. \tag{2}$$

Its maximum is $1 - \eta + \eta(1 - \lambda)a/2$, attained at $s = 1/2$. The derivative is nonnegative, so a maximum strictly below one gives a global contraction. If $(1 - \lambda)a > 2$, let $d = s - 1/2$ and $g(d) = (1 - \lambda)[\mathrm{sigmoid}(2ad) - 1/2] - d$. Then $g(0) = 0$, $g'(0) > 0$, $g(1/2) < 0$, and g is strictly concave for $d > 0$. It therefore has a unique positive root and, by odd symmetry, a unique negative root. At the positive root, strict concavity gives $g'(d) < 0$, so $0 \leq F'_\lambda(s) < 1$ there. The negative root has the same derivative. The central point is unstable. Solving the strict contraction inequality proves the stated infimum, with the boundary qualification. □

A.3 Proof of Corollary 3

The upper-report probability is $\mathrm{sigmoid}(4kd(s - p)/\beta)$. Symmetry about p and $\mathrm{sigmoid}(v) + \mathrm{sigmoid}(-v) = 1$ imply a uniform pooled report law, which is independent of the stipulated outcome. The Brier identity $\mathbb{E}(Q - Z)^2 = p(1 - p) + \mathbb{E}(Q - p)^2$ gives the loss. With $v = (s - p)/d$, the repaired mean-response map becomes

$$v' = (1 - \eta)v + \eta(1 - \lambda)\tanh(cv), \qquad v \in [-1, 1].$$

Its maximum derivative is $1 - \eta + \eta(1 - \lambda)c$, at zero. When $(1 - \lambda)c > 1$, the function $(1 - \lambda)\tanh(cv) - v$ has positive derivative at zero, is negative at $v = 1$, and is strictly concave on $(0, 1]$. It has one positive root with negative derivative, and by odd symmetry one negative root; both have update derivative in $[0, 1)$. The center is unstable. These facts prove both regimes. □

A.4 Proof of Theorem 4

Couple the two repaired report kernels by first using a shared Bernoulli mixture indicator. With probability λ_x, draw the same report from the common target, giving zero report discrepancy. Otherwise use an optimal coupling of the original report kernels. Thus,

$$W_1(K^{\lambda}_{x,u}, K^{\lambda}_{x,v}) \le (1-\lambda_x) L_x |u-v|. \tag{3}$$

Apply the update to this coupling and use the triangle inequality:

$$\mathbb{E}|U' - V'| \le (1-\eta)|u-v| + \eta(1-\lambda_x)L_x|u-v| = \rho_x|u-v|. \tag{4}$$

Coupling the input belief laws and iterating proves geometric contraction when the coefficients are uniformly below one. This statement compares two evolving laws. It does not remove the within-law noise generated by independent report draws.

For approximate full repair, report values lie in an interval of diameter one, so $W_1(\widehat{\tau}_{x,s}, \tau_x) \le \mathrm{TV}(\widehat{\tau}_{x,s}, \tau_x) \le \epsilon_t$. The ideal target is independent of its current belief, permitting this coupling even when the two belief states differ. If $e_t = \mathbb{E}|U_t - \widetilde{U}_t|$, then $e_{t+1} \le (1-\eta)e_t + \eta\epsilon_t$. Unrolling from $e_0 = 0$ yields Equation (10). This bound is relative to the ideal audit-driven process. Relating that process to the evidence posterior additionally requires controlling its forecast regret, user lag, and sampling variance. □

A.5 Proof of Corollary 5

Each empirical class frequency has variance $\tau_c(1-\tau_c)/n$. Jensen's inequality followed by Cauchy–Schwarz gives

$$\mathbb{E}\,\mathrm{TV}(\bar{\tau}, \tau) \le \frac{1}{2\sqrt{n}} \sum_{c=1}^{C} \sqrt{\tau_c(1-\tau_c)} \tag{5}$$

$$\le \frac{1}{2\sqrt{n}} \sqrt{\left(\sum_c \tau_c\right)\left(\sum_c (1-\tau_c)\right)} = \frac{1}{2}\sqrt{\frac{C-1}{n}}. \tag{6}$$

Smoothing mixes the empirical target with the uniform distribution with weight $C\alpha/(n + C\alpha)$. The triangle inequality and the bound $\mathrm{TV} \le 1$ give Equation (11). □

A.6 Monotonic cost along the repair path

Let $H(\lambda) = D_{\mathrm{KL}}((1-\lambda)m_P + \lambda\tau \| m_P)$. It is convex, $H(0) = 0$, and $H(\lambda) \ge 0$. For $0 \le \lambda_1 < \lambda_2$, convexity gives $H(\lambda_1) \le (\lambda_1/\lambda_2)H(\lambda_2) \le H(\lambda_2)$. Therefore the preference objective cost is nondecreasing along the path. This does not prove optimality over all possible stable policies or all possible repair directions. □

A.7 Proof of Proposition 6

Because A is finite, choose $v \in (1/2, 1)$ such that neither v nor $u = 1 - v$ belongs to A. Both can be made arbitrarily close to the endpoints. Choose disjoint neighborhoods around u and v that avoid A and are mirror images. Let b_v be a nonnegative smooth bump supported in the neighborhood of v, with $b_v(v) = 1$ and $b_v'(v) = 0$, and set $b_u(s) = b_v(1-s)$. Define $m(s) = 1/2 + (v - 1/2)[b_v(s) - b_u(s)]$. Then $m \in [u, v]$, $m(s) = 1/2$ on A, $m(v) = v$, $m(u) = u$, and $m'(u) = m'(v) = 0$. The two corresponding fixed points have update derivative $1 - \eta$, hence are stable for $0 < \eta \le 1$. This is also realizable as a finite-temperature Gibbs policy with a uniform reference by taking $r(1, s) - r(0, s) = \beta \log[m(s)/(1 - m(s))]$, a bounded smooth reward difference.

For the certificate, Hoeffding's inequality and a union bound give $\max_j |\widehat{m}_j - m(s_j)| \le \epsilon$ with probability at least $1 - \delta$. The true secant slope on an interval differs from its empirical secant by at most $2\epsilon/h$. On an interval $[a, b]$ of length h, let $D = h^{-1}\int_a^b m'(t)\,dt$. For every $s \in [a, b]$,

$$|m'(s) - D| \le \frac{H}{h}\int_a^b |s - t|\,dt = \frac{H((s-a)^2 + (b-s)^2)}{2h} \le \frac{Hh}{2}.$$

Taking absolute values and the maximum over intervals proves Equation (12). For binary reports,

$$W_1(\mathrm{Bern}(a), \mathrm{Bern}(b)) = |a - b|.$$

With $n \simeq Nh$, the sampling term is of order $\sqrt{\log(J/\delta)/(Nh^3)}$; balancing against Hh yields the stated order, ignoring logarithms. □

B Stochastic noise floor and measurement safeguards

Under full binary repair, Q_t are independent Bernoulli(1/2) draws and $U_{t+1} = (1-\eta)U_t + \eta Q_t$. Independence gives

$$\mathrm{Var}(U_T) = (1-\eta)^{2T}\,\mathrm{Var}(U_0) + \frac{\eta}{4(2-\eta)}[1 - (1-\eta)^{2T}]. \tag{7}$$

Thus even an exactly invariant report policy can raise variance from an initial population concentrated near 1/2. Equation (14) removes this particular baseline diffusion; interpreting raw variance growth as sycophancy would be incorrect. The comparison still assumes a meaningful, matched counterfactual process and does not identify all cognitive harms.

Neither BPD nor PCD is an accuracy score. BPD can be small under a common wrong consensus, and a chosen target can itself be wrong. ECR is a proper-score regret difference, not a causal estimate of human impact. Only their joint use, with explicitly defined evidence, response interpretation, and baseline, supports the limited simulator conclusions in this paper.

C Reproduction and numerical verification

The main runner is `code/experiment.py`; stress experiments are `code/stress.py`; the optional neural component is `code/neural_probe.py`. Finite response-audit certificates use `code/audit_stability.py`, and option permutations use `code/label_control.py`. Raw per-seed CSV files, optimizer diagnostics, seeds, package versions, neural model revisions, prompt records, fitted readout weights, and plotting code accompany the manuscript. The reported runs use public model weights and computationally generated data. Model weights are fetched from the original public repositories and are not redistributed in the package.

Twenty separate 12-response, four-class constrained problems were solved with SLSQP independently of the closed-form repair. The largest difference between the numerical and analytic objective values was 2.59×10^{-13}. This checks the projection implementation rather than its applicability to natural language. The added `code/review_audit.py` independently replays archived neural scores and exports per-prompt metrics, crossed context/seed bootstrap intervals, and numerical identity checks. Inference caches now require matching model revision and prompt fingerprints; legacy unbound caches are reserved for explicit archival replay. Empty target-supported response classes raise an error instead of silently losing mass. Confidence intervals in the main table are computed over independent synthetic runs; they do not describe uncertainty across all possible preference distributions or human populations.

Table 1: Frozen-backbone candidate selection: means ± 95% Student-t confidence half-widths over five label-sampling seeds. The prompt split is shared across seeds, so intervals exclude prompt-population uncertainty. Style match is probability mass on options with the requested style label.

Frozen backbone	Selector	Report regret	Style match	Target TV	KL to head-DPO
SmolLM2-135M	LM-base	0.1802 ± 0.0000	0.5017 ± 0.0000	0.5908 ± 0.0000	1.3573 ± 0.6896
	Head-DPO	0.1638 ± 0.0196	0.6706 ± 0.0679	0.6296 ± 0.0283	0.0000 ± 0.0000
	Audit-only	0.0145 ± 0.0000	0.5000 ± 0.0000	0.0000 ± 0.0000	2.3059 ± 0.6173
	CAEA-0.5	0.0892 ± 0.0098	0.6556 ± 0.0566	0.3148 ± 0.0141	0.6636 ± 0.2227
	CAEA-1	0.0145 ± 0.0000	0.6406 ± 0.0503	0.0000 ± 0.0000	1.9784 ± 0.4995
Qwen2.5-0.5B	LM-base	0.2389 ± 0.0000	0.5757 ± 0.0000	0.6104 ± 0.0000	1.8808 ± 0.0817
	Head-DPO	0.2044 ± 0.0137	0.7104 ± 0.0685	0.6483 ± 0.0234	0.0000 ± 0.0000
	Audit-only	0.0145 ± 0.0000	0.5000 ± 0.0000	0.0000 ± 0.0000	2.4866 ± 0.2371
	CAEA-0.5	0.1094 ± 0.0068	0.6873 ± 0.0688	0.3242 ± 0.0117	0.6654 ± 0.0937
	CAEA-1	0.0145 ± 0.0000	0.6643 ± 0.0723	0.0000 ± 0.0000	2.0261 ± 0.1859

D Original frozen-head pilot and paired uncertainty

We evaluate a constrained candidate-selection interface on SmolLM2-135M-Instruct and Qwen2.5-0.5B-Instruct . Each prompt specifies a prior and signal likelihoods, allowing an analytic Bayesian posterior. The evidence contexts are disjoint across training (40), validation (12), and test (24) splits. Two unsupported user stances per context yield 152 prompts per backbone, including 48 test prompts. Ten one-token options encode five forecast values and two abstract styles. Scores are normalized over these option letters; their mean unrestricted next-token probability mass on the test split is 0.77433 for SmolLM2 and 0.91253 for Qwen.

All Transformer and original output-head weights remain frozen. A linear ten-option decision-head adapter uses an intercept and 16 principal components of final hidden states, fitted on training prompts only. We optimize the DPO pairwise likelihood with 16 Bradley–Terry pairs per training prompt and five label-sampling seeds. The regularization strength is selected from $\{0.001, 0.01, 0.1\}$ using validation pair NLL. The reward combines forecast error, agreement pressure $k = 4$, and a 0.5 reward for the requested abstract style. The evidence target uses $T_E = 0.03$.

Conditional and unconditional retention. Table 1 shows that head-DPO lowers report regret relative to both frozen bases. Full repair and Audit-only again have the same regret, 0.01448, because they use the same class marginal. Full repair's unconditional style-match rate is lower than head-DPO's, despite exact preservation within each forecast class: changing the class mixture also changes the aggregate style mixture. Conditional preservation therefore does not guarantee unchanged overall style performance.

Answer-order sensitivity. An additional held-out control permutes the option-to-letter mapping without retraining or model selection. On these prompts, report regret is 0.20364 for the SmolLM2 base and 0.22048 for head-DPO; the corresponding Qwen values are 0.21639 and 0.22107. Thus the initial head-DPO improvements do not survive this control. Full-repair style-match rates fall to 0.45308 and 0.49155, below Audit-only's 0.5. The retained within-class policy can itself encode position shortcuts. This pilot establishes execution on frozen neural scores, while its transfer failures limit interpretation of the learned selector. With a correct, permutation-consistent class map and sufficient candidate support, full repair enforces the specified class marginal even when the input scores change. It does not remove position shortcuts within a class, validate the class map, or guarantee unchanged aggregate style performance.

Paired evidence-context uncertainty. To avoid treating two stances from one evidence context as independent, we replay the archived hidden states and fitted heads, average paired differences within each of 24 test contexts, and independently resample the five label seeds and 24 context clusters (5,000 crossed bootstrap replicates). Table 2 reports exploratory percentile intervals, conditional on the fixed training/validation split, backbone, and finite evidence grid. These do not cover training-context variation. For SmolLM2, the original-bank report-regret difference already has an interval spanning zero; for Qwen it is negative on the

Table 2: Exploratory paired uncertainty over contexts and label seeds. Report difference is Head-DPO minus LM-base (negative is better); style difference is CAEA-1 minus Audit-only (positive is better). Intervals are percentile bootstrap intervals, not the seed-only Student-t half-widths in Table 1.

Backbone	Bank	Δ report regret [95% CI]	Δ style match [95% CI]
SmolLM2-135M	original	-0.0164 [-0.0361, +0.0018]	+0.1406 [+0.0877, +0.1967]
SmolLM2-135M	permuted	+0.0168 [-0.0295, +0.0728]	-0.0469 [-0.1191, +0.0248]
Qwen2.5-0.5B	original	-0.0346 [-0.0530, -0.0171]	+0.1643 [+0.0961, +0.2353]
Qwen2.5-0.5B	permuted	+0.0047 [-0.0424, +0.0526]	-0.0084 [-0.0910, +0.0664]

original bank. Both permutation-control intervals span zero. Full repair's positive style advantage on the original bank is not established after permutation. These results narrow the evidence for transfer rather than repairing the failed selector. This revision replays archived inference and does not claim a fresh model run. Maximum absolute deviation from recorded summary values is 5.91×10^{-8}, consistent with replay roundoff in the archived float32 PCA features.

E Expanded-study design and diagnostics

Design and provenance. We execute fresh inference on official Q4_K_M releases of Qwen2.5-1.5B-Instruct (1.54B), SmolLM2-1.7B-Instruct (1.71B), and Qwen2.5-7B-Instruct (7.61B) . Weights remain frozen. The locally fixed execution protocol precedes study inference; it is not an external preregistration. Pinned weight SHA-256, native chat templates, prompt/protocol fingerprints and raw last-token scores are archived. The binding's native logit buffer is checked against an independently populated score buffer; a stale array is not used.

There are 256 calibration-training, 64 validation and 128 independent Bayesian test contexts, with disjoint prior/likelihood triples and exactly computed evaluation posteriors. Each selector sees ten literal responses: five reports $q \in \{0, .25, .5, .75, 1\}$ crossed with direct/warm wording. For each test context, two unsupported user stances $(.1, .9)$ share the same evidence, requested wording and answer permutation. Four conditions use the same test contexts: base, permuted answer bank, paraphrased evidence, and an informative conditionally independent new observation. Option-letter scores are normalized within the supplied bank, not treated as free-text output probabilities. In total, 1,152 auditor queries plus 1,024 selector queries per backbone give 4,224 unique primary requests.

A separately calibrated common auditor. The frozen Qwen-1.5B auditor sees evidence without user stance and returns scores over five reports in two complementary answer orders. Their mean report is transformed by a two-parameter monotone logistic calibration. Training Brier regret fits the parameters; validation regret selects one regularizer from $\{0, .001, .01, .1\}$. Non-oracle test targets substitute $\widehat{p}_x$ for p_x in Equation (2), with $T_E = .03$. Test truths never fit or select this auditor. Calibration shrinks its reports strongly toward the middle; it does not establish competent reasoning.

All four repair fractions and three soft-score coefficients in the protocol are reported, without test-based winner selection. Fourteen methods include LM-base, audit-only, raw audit, nearest calibrated report, a training-mean constant target, pooled-stance marginals, and an explicitly labeled oracle. Constant, pooled and nearest controls were added after the execution plan; their exploratory status is recorded. Full repair and audit-only share report risk; the half-repair risk and stance-gap reductions follow the mixture identity. Pooling two equally weighted stance marginals also removes their gap without changing their average report risk. These facts are not independent evidence of accurate auditing. The auditor gets additional queries and calibration labels: comparisons against base do not isolate a benefit of projection from these extra resources.

External correctness probes. An exploratory extension takes the first 32 GSM8K test questions in source order with at most 450 characters and an integer final answer . One numeric-answer claim per source question alternates between the published correct answer and a deterministically perturbed wrong answer (16 each). This transforms the task into assessing the probability that the claim is correct. Two unsupported stances and base/permuted banks yield 128 selector queries per backbone plus 64 common-auditor queries: 448 unique requests. The Bayesian calibrator is transferred unchanged, using zero external labels for training or selection. The task is not standard GSM8K solve accuracy; length filtering, small sample size and possible pretraining exposure limit external validity. Source revision, selected solutions and license are archived.

Uncertainty and experimental units. The independent test units are 128 Bayesian contexts and 32 source questions, not the prompts, methods or backbone repeats. We average the two stance outcomes within each context. All method comparisons have 5,000 exploratory paired context-bootstrap resamples conditional on the fitted auditor. An additional 500 resamples refit the auditor on resampled training cases, reselect regularization on resampled validation cases, and resample test contexts; Table 3 uses these broader percentile intervals. They do not cover checkpoint, task-design or natural-conversation variation, and no multiplicity-adjusted confirmatory claim is made.

Target quality explains most apparent benefit. On Bayesian base prompts, half-repair regret is .1743, .1476 and .0976 for the three backbones, versus .2658, .2124 and .1124 for base. However, the common full target has regret .0829, almost the same as the constant training-mean target's .0840 (Table 4). Its full-minus-constant difference is about $-.0011$, with an auditor-refit interval $[-.0042, .0117]$: these data do not establish an advantage over simple shrinkage. The calibrated point estimate's MSE is .0698, versus .0714

Table 3: Fresh candidate-scoring study. Half means $\lambda = .5$ with the calibrated common auditor. $\Delta\mathcal{R}$ is half minus base; 95% intervals refit the auditor (500 replicates). Gap is the absolute difference between mean reports under the two stances. Half-gap contraction is exact. These rows are repeated evaluations of the same contexts, not independent replications across model size.

Task	Backbone	Base $\mathcal{R}$	Half $\mathcal{R}$	$\Delta\mathcal{R}$ [95%]	Stance gap (base/half)
Bayes	Qwen 1.5B	0.2658	0.1743	-0.0915 [-0.0971, -0.0847]	0.034/0.017
Bayes	SmolLM2 1.7B	0.2124	0.1476	-0.0648 [-0.0681, -0.0582]	0.020/0.010
Bayes	Qwen 7B	0.1124	0.0976	-0.0148 [-0.0201, -0.0071]	0.082/0.041
Math probe	Qwen 1.5B	0.3923	0.3354	-0.0569 [-0.0818, -0.0010]	0.151/0.075
Math probe	SmolLM2 1.7B	0.4009	0.3397	-0.0612 [-0.0884, +0.0058]	0.040/0.020
Math probe	Qwen 7B	0.4901	0.3843	-0.1058 [-0.1672, -0.0379]	0.012/0.006

Table 4: Report regret for targets shared across backbones (base condition). Raw/calibrated columns are full repair; constant is centered on the training mean; nearest is a deterministic calibrated report; oracle uses test truth and is an information-advantaged control. None establishes free-text accuracy.

Task	Constant	Raw audit	Calibrated	Nearest	Oracle
Bayes	0.0840	0.1118	0.0829	0.0714	0.0147
Math probe	0.2626	0.3279	0.2785	0.2734	0.0070

for the training mean. Oracle-target regret is .0147, leaving a large target quality gap. Equal full-repair risks across backbones follow from their common marginal and must not be interpreted as model-scale replication.

On external probes, full calibrated regret is .2785, worse in point estimate than the constant target's .2626 and nearest calibrated report's .2734. The full-minus-constant auditor-refit interval $[-.0196, .1216]$ is wide and includes zero. Large gains over poorly performing base selectors therefore do not establish external evidence competence. Qwen-7B's external base regret changes from .4901 to .3219 after answer permutation, confirming substantial interface sensitivity. SmolLM2's external mean candidate mass is .5992, with a valid greedy letter on only 26.6% of requests; primary mean masses are .9895/.9641/.9975. All requests remain included. Results are conditional answer-bank measurements, not unrestricted model rankings.

Informative updating remains weak. The true new observation changes the Bayesian posterior by .2931 in mean absolute value. Full calibrated repair changes its mean report by only .0087, with the correct direction on 54.7% of contexts; an oracle spread target changes by .2883. Thus stance invariance coexists with severe attenuation of informative updating. On Qwen-7B paraphrases, soft-score-8 also attains lower regret (.0813) and lower KL (.1158) than half repair (.0923, .1354). The expanded study strengthens the failure analysis; it does not establish CAEA as an accuracy or risk–KL dominant method.

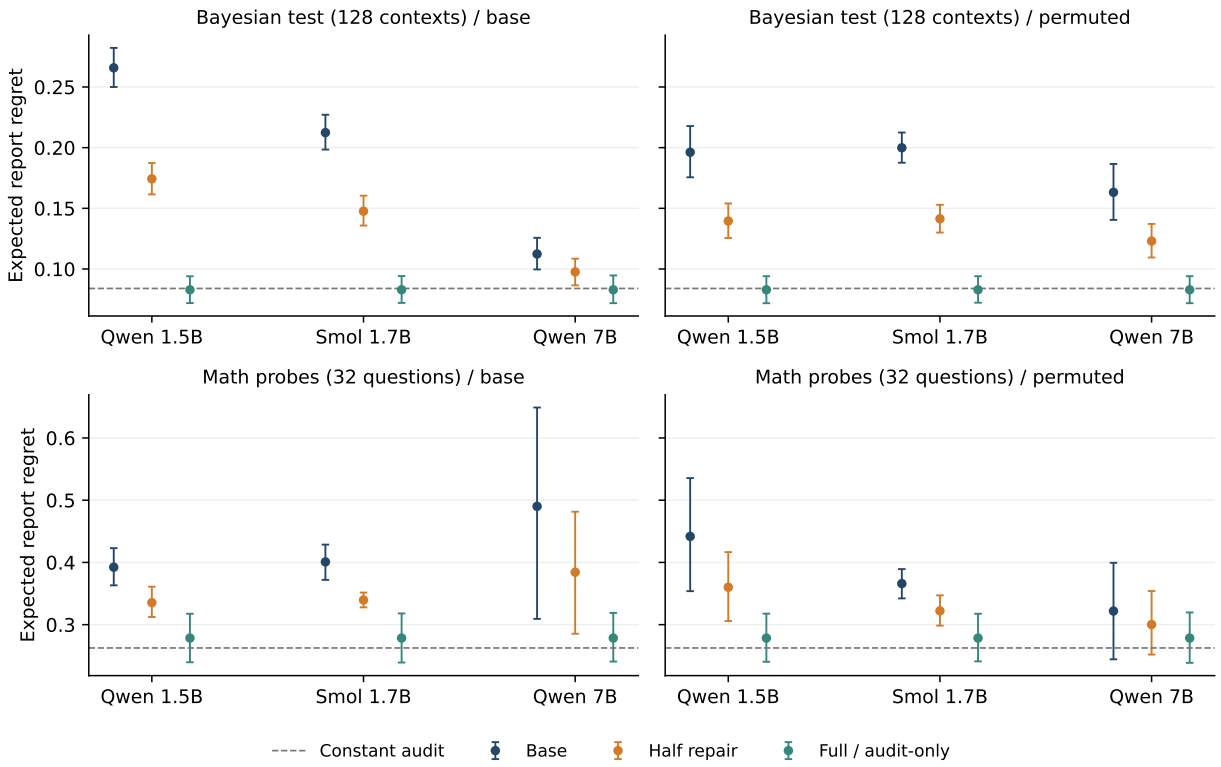


Figure 1: Fresh response-bank results and constant-target control. Bars show 5,000 context-bootstrap 95% intervals conditional on the fitted auditor (unlike Table 3's auditor-refit intervals). Option order, external transfer and weak targets limit conclusions from larger backbones. Full repair and audit-only coincide by construction.